Assignment 2: Multiple Comparisons & Permutation Tests

Behavioral Research & Statistical Methods (BRSM)

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Contents

Question 1: Bonferroni & Benjamini-Hochberg Corrections

Setup

```r
p <- c(0.0050, 0.0010, 0.0100, 0.0005, 0.0009, 0.0400, 0.0560, 0.0500,
       0.0480, 0.0130, 0.0370, 0.0430, 0.0020, 0.0250, 0.1100, 0.0700, 0.0800)

p_sorted   <- sort(p)
bonferroni <- p.adjust(p_sorted, method = "bonferroni")
bh         <- p.adjust(p_sorted, method = "BH")
```

Results Table

```r
results_q1 <- data.frame(
  Rank       = 1:length(p_sorted),
  Unadjusted = round(p_sorted, 4),
  Bonferroni = round(bonferroni, 4),
  BH         = round(bh, 4)
)
knitr::kable(results_q1, caption = "Sorted p-values with Bonferroni and BH corrections")
```

Table 1: Sorted p-values with Bonferroni and BH corrections

Rank	Unadjusted	Bonferroni	BH
1	0.0005	0.0085	0.0057
2	0.0009	0.0153	0.0057
3	0.0010	0.0170	0.0057
4	0.0020	0.0340	0.0085
5	0.0050	0.0850	0.0170
6	0.0100	0.1700	0.0283
7	0.0130	0.2210	0.0316
8	0.0250	0.4250	0.0531
9	0.0370	0.6290	0.0654
10	0.0400	0.6800	0.0654
11	0.0430	0.7310	0.0654
12	0.0480	0.8160	0.0654
13	0.0500	0.8500	0.0654
14	0.0560	0.9520	0.0680
15	0.0700	1.0000	0.0793
16	0.0800	1.0000	0.0850
17	0.1100	1.0000	0.1100

```r
cat("Bonferroni significant (adjusted p < 0.05):", sum(bonferroni < 0.05), "\n")
```

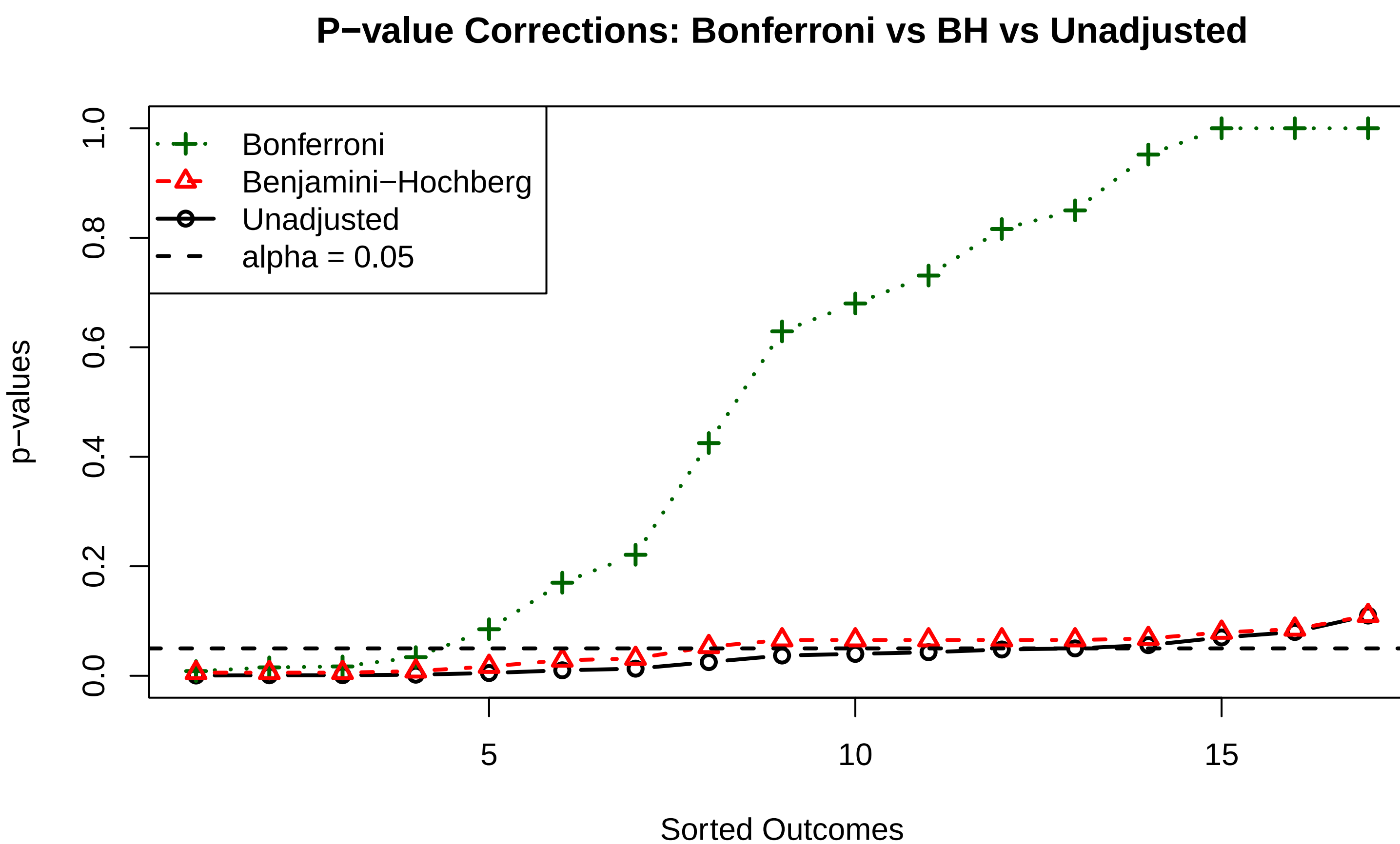
```
## Bonferroni significant (adjusted p < 0.05): 4
```

```r
cat("BH significant (adjusted p < 0.05):",         sum(bh < 0.05), "\n")
```

```
## BH significant (adjusted p < 0.05): 7
```

Plot

```r
plot(1:length(p_sorted), p_sorted,
     type = "b", col = "black", lwd = 2, pch = 1,
     ylim = c(0, max(bonferroni, na.rm = TRUE)),
     xlab = "Sorted Outcomes", ylab = "p-values",
     main = "P-value Corrections: Bonferroni vs BH vs Unadjusted")
lines(1:length(bonferroni), bonferroni,
      type = "b", col = "darkgreen", lwd = 2, lty = 3, pch = 3)
lines(1:length(bh), bh,
      type = "b", col = "red", lwd = 2, lty = 2, pch = 2)
abline(h = 0.05, col = "black", lwd = 2, lty = 2)
legend("topleft",
       legend = c("Bonferroni", "Benjamini-Hochberg", "Unadjusted", "alpha = 0.05"),
       col    = c("darkgreen", "red", "black", "black"),
       lty    = c(3, 2, 1, 2), pch = c(3, 2, 1, NA),
       lwd = 2, bg = "white")
```



Interpretation

Bonferroni correction is more conservative: it adjusts each p-value by multiplying by the total number of tests ($m = 17$), controlling the **family-wise error rate (FWER)**. This results in only **4 significant tests**.

Benjamini-Hochberg (BH) controls the **False Discovery Rate (FDR)** instead — a less stringent criterion that allows a small expected proportion of false discoveries among all rejected hypotheses. Accordingly, BH declares **7 tests significant**, making it the less conservative approach.

The results support the claim that Bonferroni is more conservative: the Bonferroni-adjusted p-values are substantially higher than BH-adjusted ones, and Bonferroni rejects fewer hypotheses at $= 0.05$.

Question 2: Permutation Test — Mosquito Attractiveness

Data

The data are from Lefèvre et al. (2010), which examined whether beer consumption increases human attractiveness to mosquitoes.

- **Group 1 (Beer):** $n = 25$
- **Group 2 (Water):** $n = 18$

```r
# Read from the local CSV in the assignment folder
mosquito_file <- "assignemnt_2_data_BRSM_Results Visualization(Mosquito).csv"

if (file.exists(mosquito_file)) {
  mosq  <- read.csv(mosquito_file, fileEncoding = "UTF-8-BOM")
  mosq  <- mosq[!is.na(mosq$No_of_Mosquitoes), ]
  beer  <- mosq$No_of_Mosquitoes[mosq$Group == "Beer"]
  water <- mosq$No_of_Mosquitoes[mosq$Group == "Water"]
} else {
  # Fallback: published values from Lefevre et al. (2010)
  beer  <- c(27,19,20,20,23,17,21,24,31,26,28,20,27,19,25,31,24,28,24,29,21,21,18,27,20)
  water <- c(21,19,13,22,15,22,15,22,20,12,24,24,21,19,18,16,23,20)
}

combined <- c(beer, water)
n_beer   <- length(beer)
n_water  <- length(water)
n_total  <- length(combined)
n_iter   <- 10000

cat("Beer  group: n =", n_beer,  "| median =", median(beer),  "| mean =", round(mean(beer), 2),  "\n")
```

```
## Beer  group: n = 25 | median = 24 | mean = 23.6
```

```r
cat("Water group: n =", n_water, "| median =", median(water), "| mean =", round(mean(water), 2), "\n")
```

```
## Water group: n = 18 | median = 20 | mean = 19.22
```

Part (a): Difference in Medians — Directional HA

H : median(beer) median(water)
H (directional): median(beer) > median(water)

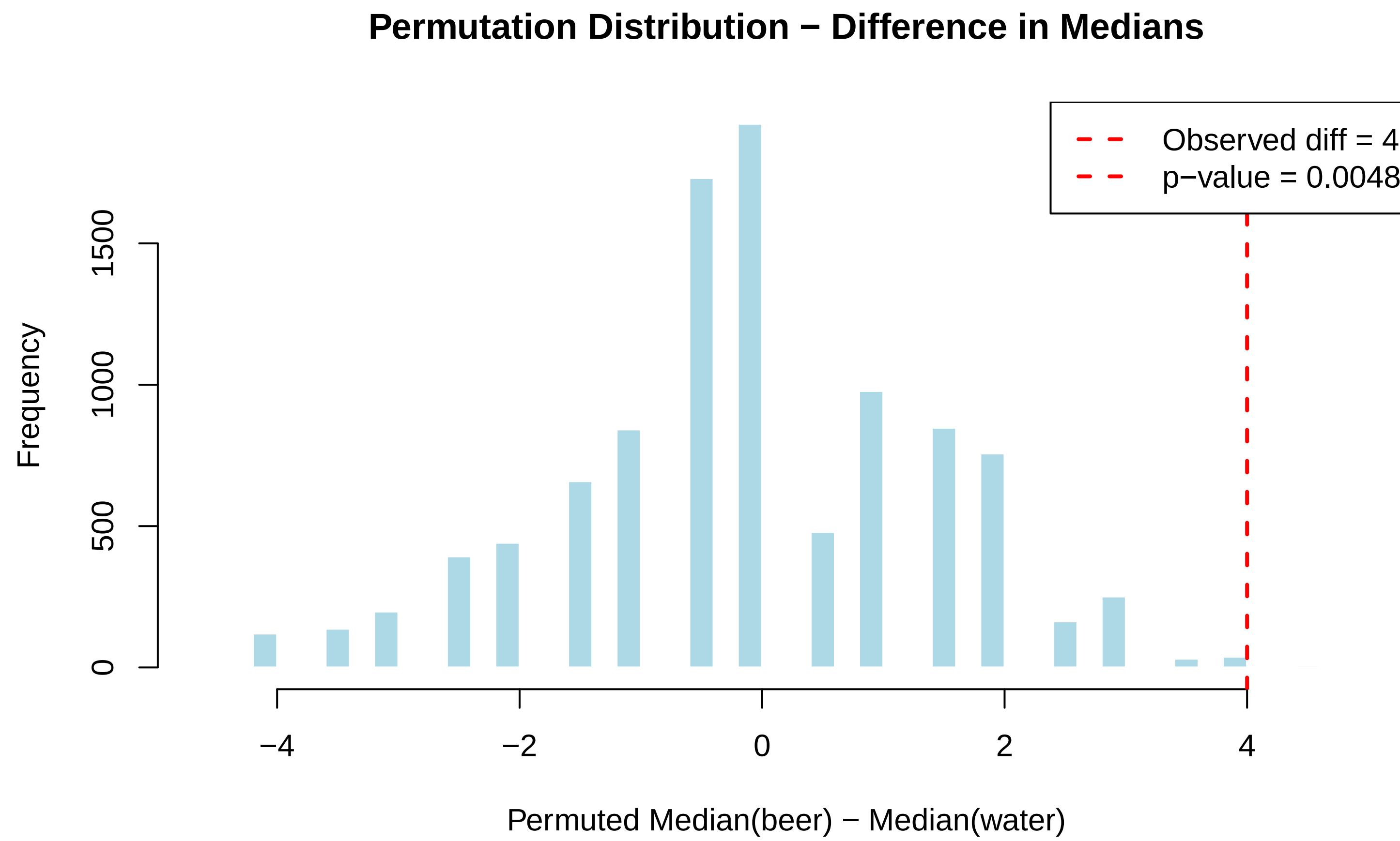
```
obs_diff_median <- median(beer) - median(water)
cat("Observed difference in medians:", obs_diff_median, "\n")
```

```
## Observed difference in medians: 4
```

```
perm_diffs <- numeric(n_iter)
for (i in 1:n_iter) {
  shuf          <- sample(combined, n_total, replace = FALSE)
  perm_diffs[i] <- median(shuf[1:n_beer]) - median(shuf[(n_beer + 1):n_total])
}

p_val_2a <- mean(perm_diffs >= obs_diff_median)
cat("One-tailed p-value (median diff):", p_val_2a, "\n")
```

```
## One-tailed p-value (median diff): 0.0048
```

```
hist(perm_diffs, breaks = 40, col = "lightblue", border = "white",
     main = "Permutation Distribution - Difference in Medians",
     xlab = "Permuted Median(beer) - Median(water)")
abline(v = obs_diff_median, col = "red", lwd = 2, lty = 2)
legend("topright",
       legend = c(paste("Observed diff =", obs_diff_median),
                  paste("p-value =", p_val_2a)),
       col = "red", lty = 2, lwd = 2, bg = "white")
```



Interpretation: The observed difference in medians is **4**. The one-tailed permutation p-value is **0.0048**, which is less than 0.05. We reject H and conclude that beer consumption significantly increases mosquito attractiveness.

Part (b): t-Statistic — Directional HA

H : mean(beer) mean(water)
H (directional): mean(beer) > mean(water)

```
obs_t <- t.test(beer, water, var.equal = FALSE)$statistic
cat("Observed t-statistic:", round(obs_t, 4), "\n")
```

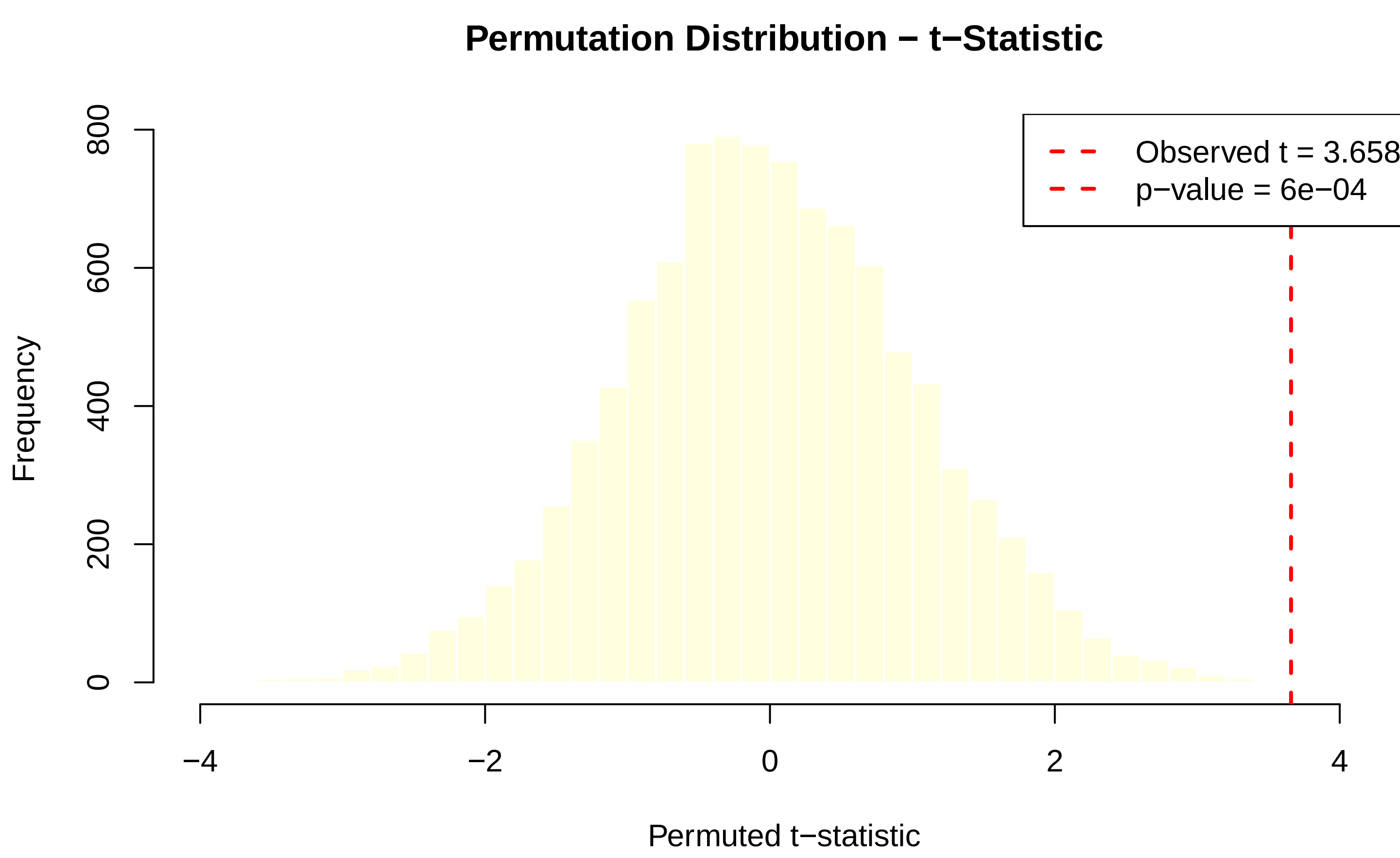
```
## Observed t-statistic: 3.6582
```

```
perm_t <- numeric(n_iter)
for (i in 1:n_iter) {
  shuf      <- sample(combined, n_total, replace = FALSE)
  g1        <- shuf[1:n_beer]
  g2        <- shuf[(n_beer + 1):n_total]
  perm_t[i] <- t.test(g1, g2, var.equal = FALSE)$statistic
}
```

```r
p_val_2b <- mean(perm_t >= obs_t)
cat("One-tailed p-value (t-stat):", p_val_2b, "\n")
```

```
## One-tailed p-value (t-stat): 6e-04
```

```r
hist(perm_t, breaks = 40, col = "lightyellow", border = "white",
     main = "Permutation Distribution - t-Statistic",
     xlab = "Permuted t-statistic")
abline(v = obs_t, col = "red", lwd = 2, lty = 2)
legend("topright",
       legend = c(paste("Observed t =", round(obs_t, 3)),
                  paste("p-value =", p_val_2b)),
       col = "red", lty = 2, lwd = 2, bg = "white")
```



Interpretation: The observed t-statistic is **3.6582**. The one-tailed permutation p-value is 6×10^{-4}, strongly significant. The t-statistic gives a more powerful test here as it uses all information in the data (means and variances), not just ranks.

Part (c): Non-directional H (Two-tailed)

H : No difference between groups
H (non-directional): There is *some* difference (either direction)

```
p_val_2a_two <- mean(abs(perm_diffs) >= abs(obs_diff_median))
p_val_2b_two <- mean(abs(perm_t)     >= abs(obs_t))

cat("Two-tailed p-value (median diff):", p_val_2a_two, "\n")
```

```
## Two-tailed p-value (median diff): 0.0169
```

```
cat("Two-tailed p-value (t-stat):     ", p_val_2b_two, "\n")
```

```
## Two-tailed p-value (t-stat):      7e-04
```

Interpretation: Under a non-directional H , both test statistics remain significant ($p < 0.05$). The two-tailed p-values are approximately double the one-tailed values (as expected), but the evidence is still strong enough to reject H in both cases.

Question 3: Permutation Test — Correlation of IQ and Test Scores

Load Data

```
library(readxl)
iq_data  <- read_excel("assignment_2_IQ.xlsx")
colnames(iq_data) <- c("ID", "GPA", "IQ", "GENDER", "TESTSCORE")
iq_vals  <- iq_data$IQ
ts_vals  <- iq_data$TESTSCORE

cat("n =", nrow(iq_data), "\n")
```

```
## n = 78
```

```
cat("IQ range:        ", range(iq_vals),  "\n")
```

```
## IQ range:         72 136
```

```
cat("TestScore range: ", range(ts_vals), "\n")
```

```
## TestScore range:  20 80
```

Observed Correlation

```r
obs_cor <- cor(iq_vals, ts_vals, method = "pearson")
cat("Observed Pearson r:", round(obs_cor, 4), "\n")
```

```
## Observed Pearson r: 0.4931
```

Permutation Test

H : No correlation between IQ and test scores (= 0)
H (non-directional): There is a correlation (0)

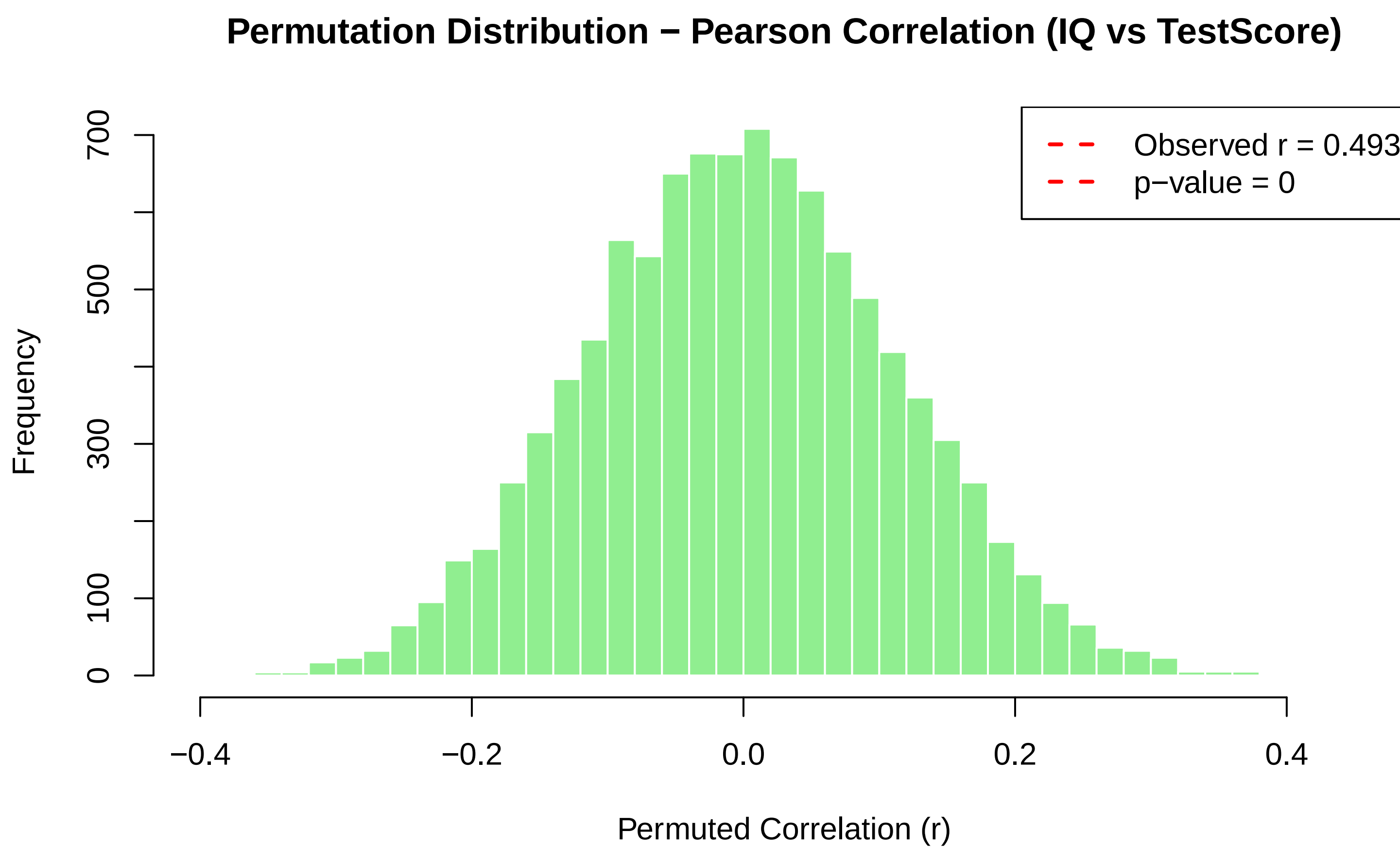
```r
perm_cor <- numeric(n_iter)
for (i in 1:n_iter) {
  perm_cor[i] <- cor(iq_vals, sample(ts_vals), method = "pearson")
}

p_val_3 <- mean(abs(perm_cor) >= abs(obs_cor))
cat("Two-tailed permutation p-value:", p_val_3, "\n")
```

```
## Two-tailed permutation p-value: 0
```

```r
cat("Decision:", ifelse(p_val_3 < 0.05,
                        "REJECT H0 - significant correlation exists",
                        "FAIL TO REJECT H0"), "\n")
```

```
## Decision: REJECT H0 - significant correlation exists
```

```r
hist(perm_cor, breaks = 40, col = "lightgreen", border = "white",
     main = "Permutation Distribution - Pearson Correlation (IQ vs TestScore)",
     xlab = "Permuted Correlation (r)")
abline(v =  obs_cor, col = "red", lwd = 2, lty = 2)
abline(v = -obs_cor, col = "red", lwd = 2, lty = 2)
legend("topright",
       legend = c(paste("Observed r =", round(obs_cor, 3)),
                  paste("p-value =", p_val_3)),
       col = "red", lty = 2, lwd = 2, bg = "white")
```



Interpretation

The observed Pearson correlation between IQ and test scores is **r = 0.4931**, indicating a moderate positive relationship. The permutation p-value is **0** — none of the 10,000 random permutations produced a correlation as extreme as the observed one.

Conclusion: We **reject H** . There is a statistically significant positive correlation between IQ and test scores ($p < 0.05$). Students with higher IQ scores tend to obtain higher test scores in this sample.